Suvarna Ramu
8401 Rosebay ln, Manassas, Virginia| **suvarnaadityan@gmail.com** | www.linkedin.com/in/suvarnaramu

SUMMARY OF QUALIFICATIONS

- Computer Engineering graduate with expertise in frontend development (HTML, JavaScript), responsive design, and software testing principles.
- Hands-on experience in Android development, debugging, and enhancing UI/UX.
- Strong foundation in algorithms, data structures, software engineering principles, design patterns, and architectural styles.
- Skilled in AI/ML concepts including search algorithms, probabilistic reasoning, and neural networks.
- Gained knowledge in Tableau with hands-on experience in data visualization and interactive dashboards.

LANGUAGES AND TECHNOLOGIES

- **Languages:** HTML, CSS, C, C++, Python, Java
- **Frameworks & Tools:** Git, Eclipse, NumPy, pandas, Matplotlib, Pytorch, TensorFlow, Keras, Scikit Learn.
- **Database Skills:** SQL, experience with Python for data querying
- **Operating Systems**: Windows, Linux.

WORK EXPERIENCE

Software Developer| SSSN Software Pvt. Ltd (2022-2023)

- Engineered a messaging application for Android, enhancing system architecture using Eclipse software.
- Achieved a 20% improvement in system performance by optimizing code and resolving data insufficiencies.
- Collaborated with UX/UI designers to implement responsive web designs.
- Increased user satisfaction by 25% through design enhancements.
- Developed and executed a unit testing framework that improved code reliability, resulting in the identification of previously undetected bugs before deployment, enhancing application stability for end-users.

Software Test Engineer | Qualcomm (2021-2022)

- Led quality assurance efforts through manual and automated testing, enhancing product reliability and performance.
- Improved product quality by 30% by identifying and resolving connection failures.
- Tested boards and connections, providing new part qualification support for memory chips using PINE and RDX boards.
- Reduced failure rates by 15% through thorough testing and analysis.

PROJECTS

Credit Card Fraud Detection using Deep Learning Techniques

- Developed a backend system with a web interface to visualize fraud detection metrics, achieving 96% accuracy using Python and deep learning algorithms.
- Employed unsupervised learning algorithms (Auto-encoder and RBM) to identify anomalous patterns, predicting fraudulent transactions with 99% accuracy on the European dataset and 96% on the Australian dataset.

Social Network Analysis Using Centrality Measures

- Created a web application for visualizing social network data using JavaScript libraries for interactive data representation. Enhanced user engagement by 40% through intuitive design.
- Conducted a comprehensive evaluation of social network datasets, focusing on four centrality measures; analysis led to refined methodologies adopted by the analytics team, ultimately improving data processing time by 15 hours monthly.

CERTIFICATIONS

- **The Complete Tableau Bootcamp for Aspiring Data Scientists (Coursera)**
  Gained strong knowledge in data visualization, dashboard creation, and storytelling with data and worked with different kinds of charts and maps. Had hands-on experience designing interactive dashboards, applying filters, and analyzing business data to extract insights.

PUBLICATIONS

- **Reviewof Feature Selection Methods and Semi Supervised Feature Selection Algorithms for Classification.**
  International Journal of Software Computing and Testing · Aug 7, 2020
  - Presents an extensive preliminary understanding about feature selection surveys recent semi-supervised feature selection methods.
- **Context Aware Physical Activity Recognition Using Social Objects**
  International Journal of Computer Science and Programming Language, July 13, 2020
  - Proposed an application that classifies physical activities (sitting, standing, walking, etc.) using smartphones without external servers, providing real-time activity notifications.
- **Credit Card Fraud Detection Using Federated Learning Techniques**
  International Journal of Scientific Research in Science, Engineering and Technology, June 11, 2020
  - Explored unsupervised learning algorithms (Auto-encoder, RBM) within a federated learning framework to predict fraudulent credit card usage.
- **Critical Care Monitoring with Event Prioritization Using IoT**
  International Journal of Scientific Research in Computer Science, Engineering and Information Technology, March 31, 2018
  - Proposed a smart healthcare system utilizing IoT to monitor critical patient parameters, prioritizing data for medical professionals based on urgency.

EDUCATION

M.S. Computer Science and Engineering, 3.54/4.0, *George Mason University, Fairfax, USA.*
M.E. Computer Science and Engineering, 3.6/4.0, *AnnaUniversity,India*.